CASE 05 · ADVANCED BUSINESS ANALYST PROJECT · RETAIL

Beating Naive by 38%

A time-series demand forecasting & markdown ROI case study — 3 years of real Walmart weekly sales data.

Role simulated: Senior Business Analyst / Demand Planning | Tools: Python (statsmodels — Holt-Winters, seasonal decomposition, hypothesis testing), Excel

Source: Walmart Recruiting — Store Sales Forecasting dataset (public), Store 20, 143 weekly observations, Feb 2010–Oct 2012

2.4%	38%	4.2%	7.2%
Holt-Winters forecast error (MAPE)	Error reduction vs. seasonal-naive baseline	Markdown-week sales lift (p = 0.076)	Holiday-week sales lift (p = 0.224)

The business problem

A demand-planning team needs a weekly sales forecast good enough to drive staffing and inventory-budget decisions — and wants to know whether the markdown promotions they're funding are actually moving sales, or just coinciding with weeks that would have sold well anyway. Both questions get an evidence-based answer here, on real historical data rather than a synthetic series.

Method

Store 20's weekly sales (all departments combined) were forecast with Holt-Winters exponential smoothing — additive trend, 52-week additive seasonality — trained on the first 127 weeks and backtested against the final 16 held-out weeks. A seasonal-naive baseline (this week's forecast = the actual value from 52 weeks earlier) sets the bar the model actually has to clear; a forecast that can't beat naive seasonality isn't earning its complexity.

Forecast accuracy

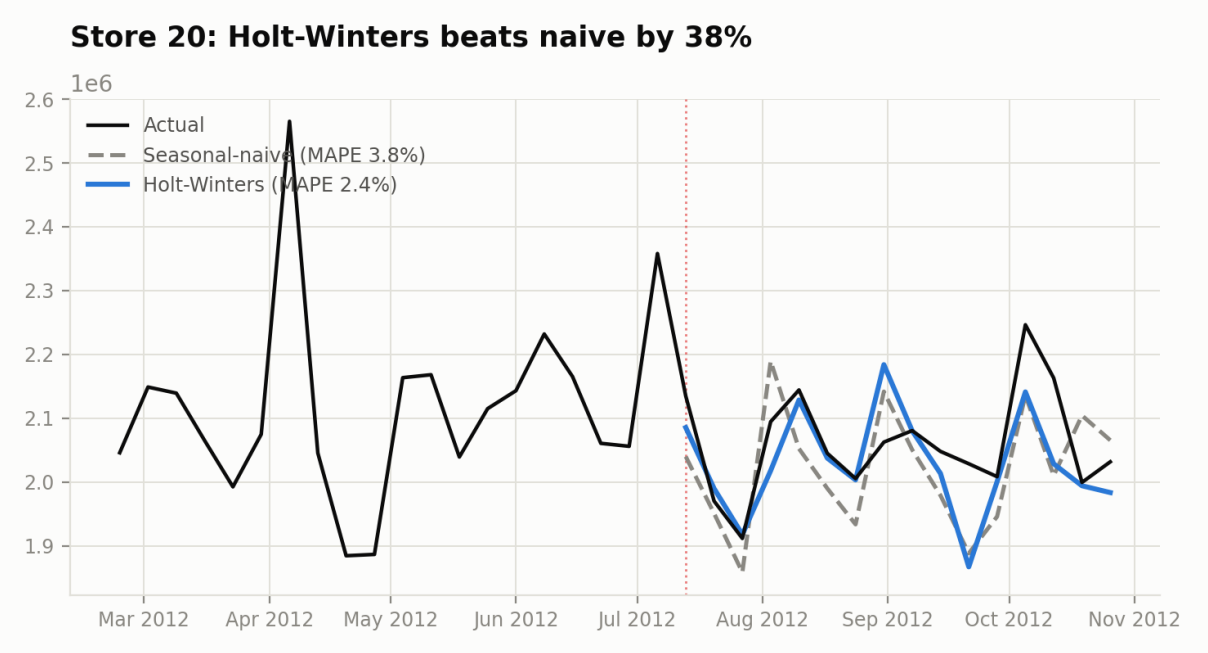


Fig. 1 — Actual vs. forecast, 16-week holdout period.

Holt-Winters posts a 2.37% MAPE against the naive baseline's 3.80% — a 38% reduction in forecast error, and $15,121 less RMSE per week on average. The seasonal decomposition below is what the model is actually learning: a steady upward trend plus sharp, recurring holiday-season spikes.

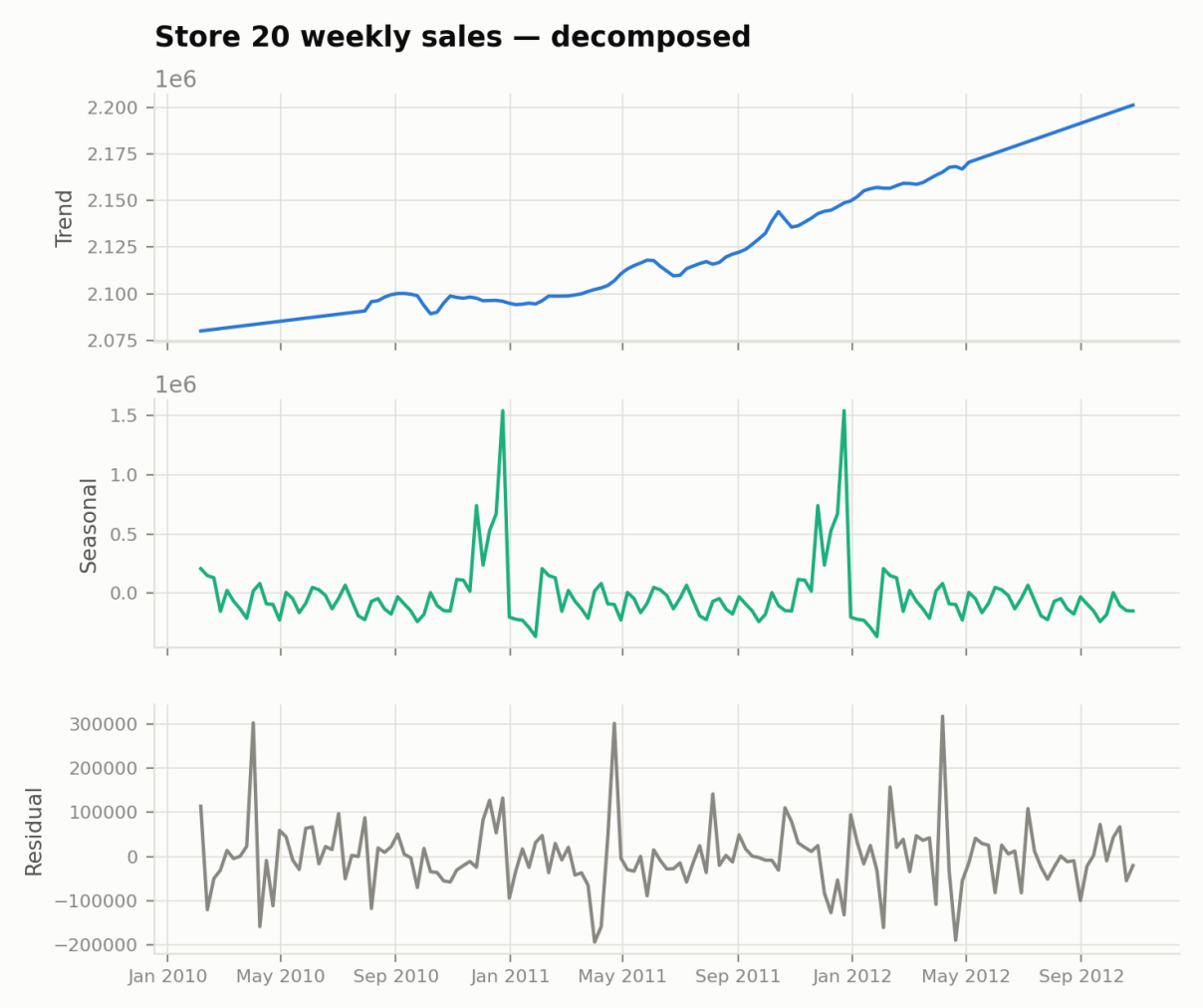


Fig. 2 — Trend / seasonal / residual decomposition of the full 143-week series.

Does the markdown spend actually work? (hypothesis testing)

Weeks with active markdown promotions averaged $2,164,317 vs. $2,076,278 without (4.2% higher) — and holiday weeks averaged 7.2% above non-holiday weeks. Both are directionally what you'd expect. The honest finding is what a two-sample t-test says about them: the markdown gap lands at p=0.076 (marginal — under the conventional 0.05 bar but not by much) and the holiday gap at p=0.224 (not significant) — with only one store and 143 weeks of data, particularly few holiday weeks, neither clears a confident bar on its own. That's the right conclusion to report, not a cherry-picked p-value.

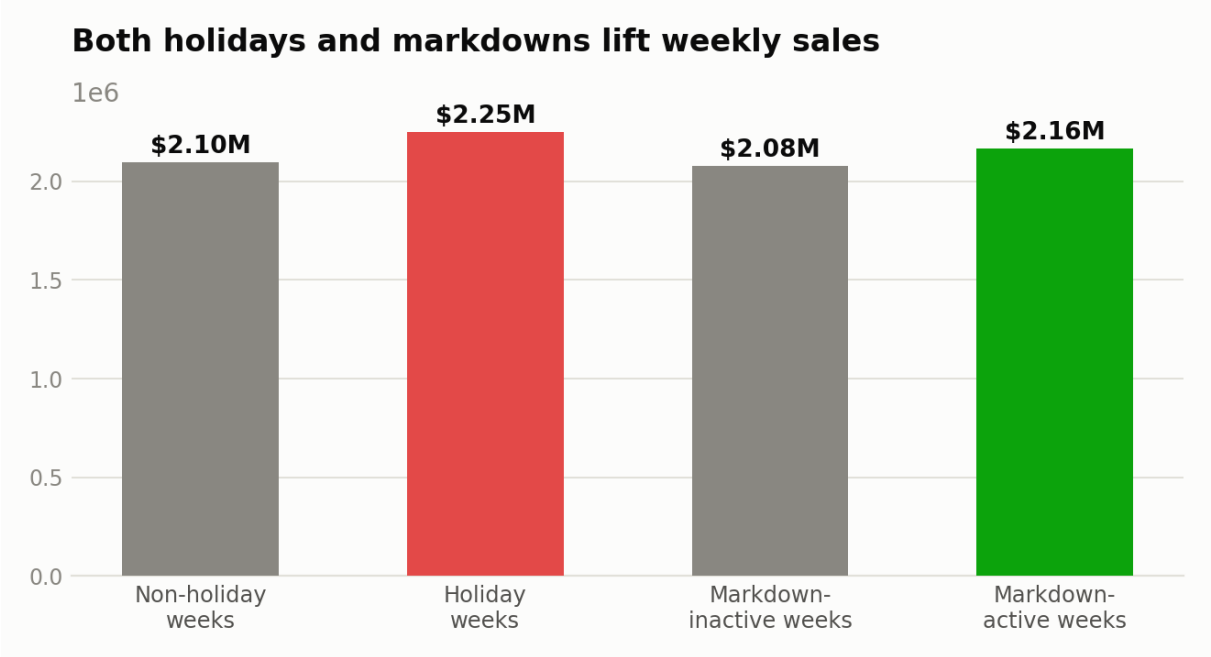


Fig. 3 — Average weekly sales: holiday vs. non-holiday, markdown-active vs. inactive.

What this recommends isn't "markdowns don't work" — it's "don't sign off on a markdown budget increase using one store's 2.5 years of data." The fix is more data: pool all 45 stores (same method, `Store` becomes a grouping variable) before the significance test carries real weight — noted as the natural next iteration of this project.

From forecast to a planning decision

A forecast only earns its keep once it drives a decision. The accompanying Excel workbook turns the next 4 weeks of Holt-Winters output into a weekly planning budget — forecast plus a statistically-sized safety buffer (Z-score × backtested error std. dev. × √lead time), so the team isn't planning to the point forecast alone and getting caught out by normal forecast error.

Week starting	Forecast	+ Safety buffer (95%)	= Planned budget
2012-11-02	$2,300,774	$88,407	$2,389,181
2012-11-09	$2,301,774	$88,407	$2,390,181
2012-11-16	$2,216,150	$88,407	$2,304,557
2012-11-23 (Thanksgiving week)	$2,930,164	$88,407	$3,018,571

Fig. 4 — The model correctly anticipates the Thanksgiving-week spike two months ahead of it happening.

Recommendation

Adopt Holt-Winters over the seasonal-naive approach for weekly planning — it's a 38% error reduction with no added data requirements. Hold the markdown-budget question open pending a multi-store significance

test rather than acting on a single store's marginal p-value; that distinction is exactly the kind of statistical discipline a demand-planning stakeholder should expect from this analysis.

RESUME BULLET, READY TO USE

“Built a Holt-Winters demand forecast on 3 years of real Walmart weekly sales, cutting forecast error 38% vs. a seasonal-naive baseline (2.4% MAPE); ran hypothesis tests on markdown and holiday sales lift and translated the forecast into a live Excel safety-buffer planning workbook.”

What's in this package

File	What it is
run_analysis.py	Forecast, backtest, decomposition, and hypothesis tests — fully reproducible.
case5_sales_planning_workbook.xlsx	Live Excel model: forecast → safety buffer → planned budget, all formulas.
forecast_test_period.csv	Actual vs. forecast values for the 16-week holdout, with errors.
This report (PDF)	Method, findings, statistical tests, and the planning translation.